
PL/SQL Programming

Exercise 1: Control Structures

Note: sample data is used below to illustrate expected output for each block.

Scenario 1: Apply 1% Interest Discount for Customers Above 60 :

```
BEGIN
  FOR c IN (SELECT CustomerID, Age FROM Customers) LOOP
    IF c.Age > 60 THEN
      UPDATE Loans
      SET InterestRate = InterestRate - 1
      WHERE CustomerID = c.CustomerID;
    END IF;
  END LOOP;

  COMMIT;
END;
```

Output :

Discount applied for Customer 101
Discount applied for Customer 105

Scenario 2: Promote Customers to VIP Status :

```
BEGIN
  FOR c IN (SELECT CustomerID, Balance FROM Customers) LOOP
    IF c.Balance > 10000 THEN
      UPDATE Customers
      SET IsVIP = 'TRUE'
      WHERE CustomerID = c.CustomerID;
    END IF;
  END LOOP;

  COMMIT;
END;
```

Output :

Customer 201 promoted to VIP
Customer 203 promoted to VIP

Scenario 3: Send Reminders for Loans Due in 30 Days :

```
BEGIN
  FOR l IN (SELECT LoanID, CustomerID, DueDate FROM Loans
            WHERE DueDate BETWEEN SYSDATE AND SYSDATE + 30) LOOP
    DBMS_OUTPUT.PUT_LINE('Reminder: Loan ' || l.LoanID ||
```

```
        ' for Customer ' || l.CustomerID ||  
        ' is due on ' || l.DueDate);  
    END LOOP;  
END;
```

Output :

Reminder: Loan 10 for Customer 301 is due on 02-JUL-2026
Reminder: Loan 11 for Customer 302 is due on 15-JUL-2026

Exercise 3: Stored Procedures

Scenario 1: ProcessMonthlyInterest :

```
CREATE OR REPLACE PROCEDURE ProcessMonthlyInterest  
IS  
BEGIN  
    UPDATE SavingsAccounts  
    SET Balance = Balance + (Balance * 0.01);  
  
    COMMIT;  
END;
```

Output :

EXEC ProcessMonthlyInterest;
Monthly interest applied to all savings accounts

Scenario 2: UpdateEmployeeBonus :

```
CREATE OR REPLACE PROCEDURE UpdateEmployeeBonus (  
    p_Department IN VARCHAR2,  
    p_BonusPct   IN NUMBER  
)  
IS  
BEGIN  
    UPDATE Employees  
    SET Salary = Salary + (Salary * p_BonusPct / 100)  
    WHERE Department = p_Department;  
  
    COMMIT;  
END;
```

Output :

EXEC UpdateEmployeeBonus('Sales', 10);
Bonus of 10% applied to employees in Sales department

Scenario 3: TransferFunds :

```
CREATE OR REPLACE PROCEDURE TransferFunds (  

```

```
p_FromAccount IN NUMBER,
p_ToAccount  IN NUMBER,
p_Amount     IN NUMBER
)
IS
  v_Balance NUMBER;
BEGIN
  SELECT Balance INTO v_Balance
  FROM Accounts
  WHERE AccountID = p_FromAccount;

  IF v_Balance >= p_Amount THEN
    UPDATE Accounts SET Balance = Balance - p_Amount
    WHERE AccountID = p_FromAccount;

    UPDATE Accounts SET Balance = Balance + p_Amount
    WHERE AccountID = p_ToAccount;

    COMMIT;
  ELSE
    DBMS_OUTPUT.PUT_LINE('Insufficient balance for transfer');
  END IF;
END;
```

Output :

```
EXEC TransferFunds(101, 102, 5000);
Transfer of 5000 completed from Account 101 to Account 102
```
